

Reducing New-Customer Churn

Technical Report — Capstone Business Analysis, Week 8

This report documents the complete methodology, statistical validation, predictive model, and customer segmentation behind the executive summary's recommendations. The reproducible code is in **capstone_analysis.ipynb**.

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1. Project Overview & Business Problem

A subscription business needs to reduce customer churn: every lost customer represents lost recurring revenue and a wasted acquisition cost. This analysis answers: who is at risk of churning, why, and what should the business do about it?

Objectives

- Quantify the current churn rate and its business cost.
- Identify statistically-validated churn drivers (not just visual correlations).
- Build a model that flags at-risk customers before they leave.
- Segment the customer base to target retention spend efficiently.
- Translate findings into a phased implementation plan.

Success Metrics

A validated churn-driver list; a model with recall ≥ 0.90 on churners (missing an at-risk customer costs more than a false alarm); a segmentation identifying disproportionate-risk groups; and a written implementation plan with owners and trackable success metrics.

2. Dataset Selection & Data Quality

Three datasets were available. The assignment requires a minimum of 500 rows; only **customer_churn.csv** qualifies.

Dataset	Rows	Selected?
sales_data.csv	100	No — below the 500-row minimum
house_prices.csv	300	No — below the 500-row minimum
customer_churn.csv	500	Yes

customer_churn.csv (500 rows): CustomerID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, Churn. Data quality checks found: zero missing values, zero duplicate rows, zero duplicate customer IDs, no implausible numeric ranges, and consistent categorical labels. No repair was required; a cleaning pipeline was still built and documented (feature engineering: tenure buckets, paperless-billing flag, average monthly spend-to-date) so it is ready for messier future data refreshes.

3. Exploratory Data Analysis

Overall churn rate is 10.6% (53 of 500 customers).

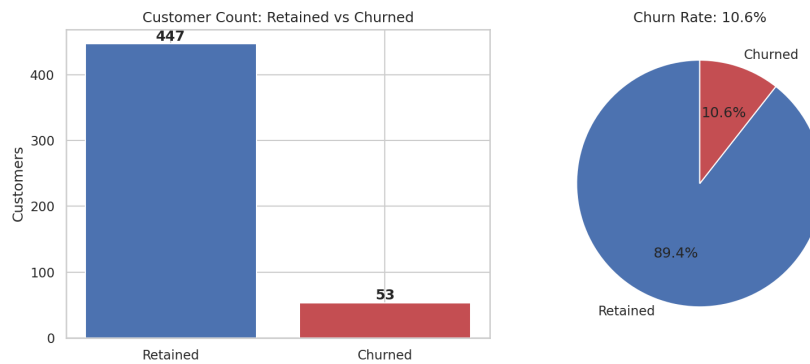


Figure 1. Churn overview.

The most important pattern in the dataset appears when churn is examined against tenure: **100% of churned customers have 11 months of tenure or less**, and churn beyond 12 months is effectively zero (Figure 2). This reframes the business problem from general churn prediction to first-year retention specifically.

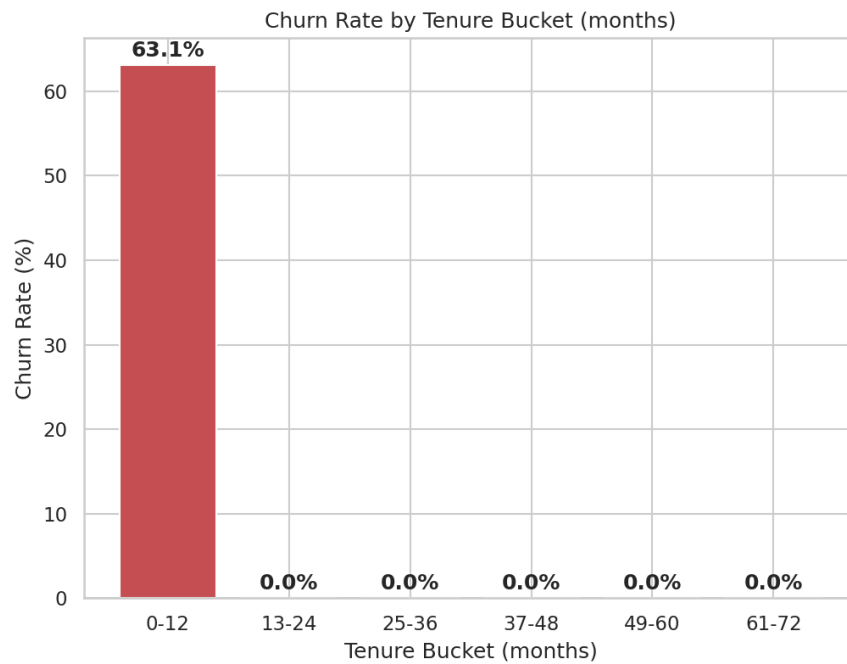


Figure 2. The "First-Year Cliff."

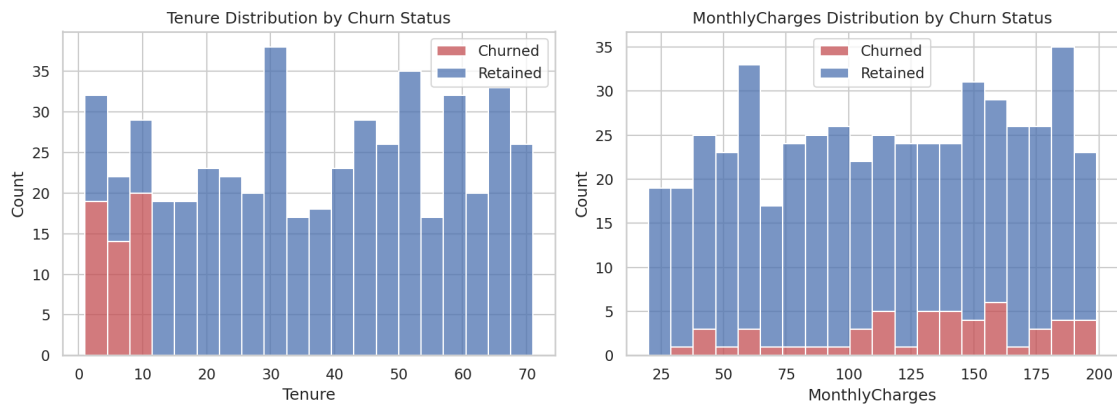


Figure 3. Tenure and MonthlyCharges distributions by churn status.

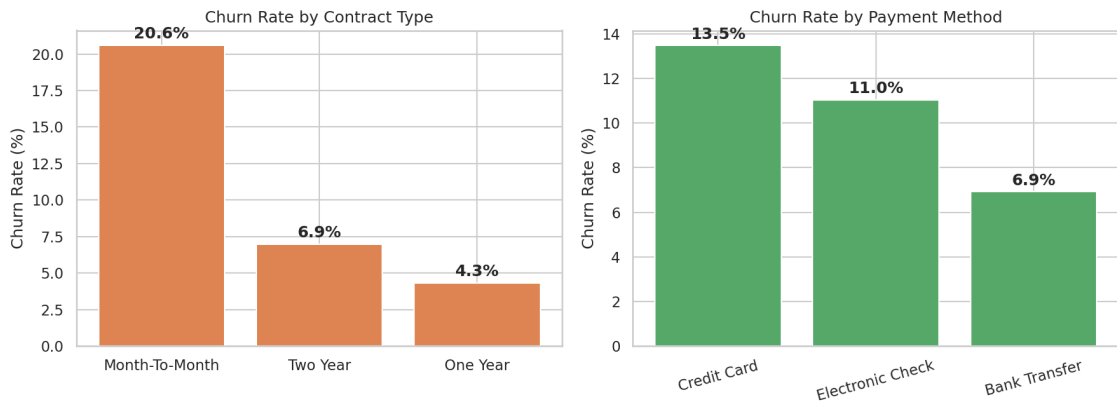


Figure 4. Churn rate by contract type and payment method.

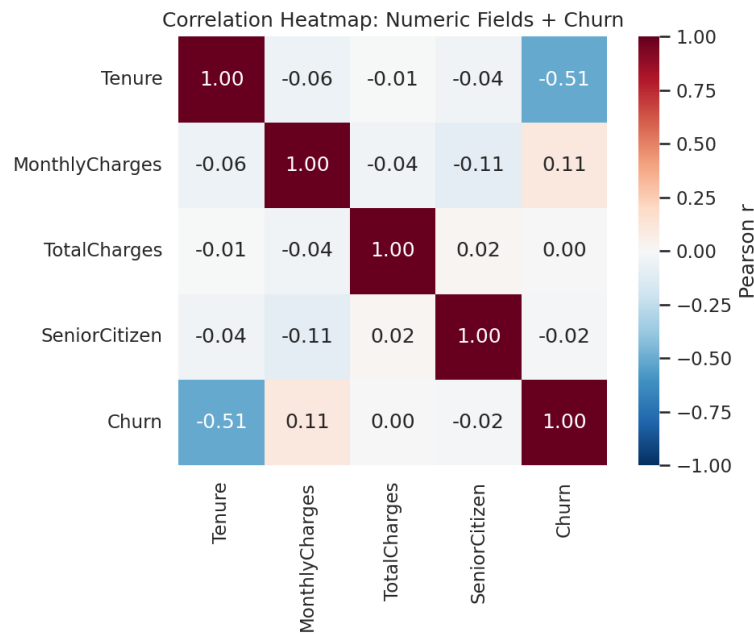


Figure 5. Correlation heatmap of numeric fields and churn.

4. Hypothesis Testing

Five hypothesis tests validate which EDA patterns are statistically real, at $\alpha=0.05$. Levene's test checked the equal-variance assumption before each t-test.

Test	p-value	Result
Contract type vs Churn (Chi-square)	<0.001	Significant
Payment method vs Churn (Chi-square)	0.145	Not significant
MonthlyCharges: Churned vs Retained (t-test)	0.010	Significant
Tenure: Churned vs Retained (t-test)	<0.001	Significant
Senior citizen status vs Churn (Chi-square)	0.795	Not significant

Contract type and tenure are confirmed, statistically significant churn drivers. Monthly charges show a smaller but still significant effect (churned customers pay more per month on average). Payment method and senior-citizen status are not significant at this sample size — reported honestly rather than omitted.

5. Predictive Modeling

A logistic regression was trained on a 75/25 train/test split (stratified by churn), using `class_weight='balanced'` because churn is rare (10.6%) and the business priority is recall — a missed churner costs more than an unnecessary retention email.

Metric	Value
Accuracy	0.944
Precision	0.650
Recall	1.000
F1 score	0.788
ROC AUC	0.987

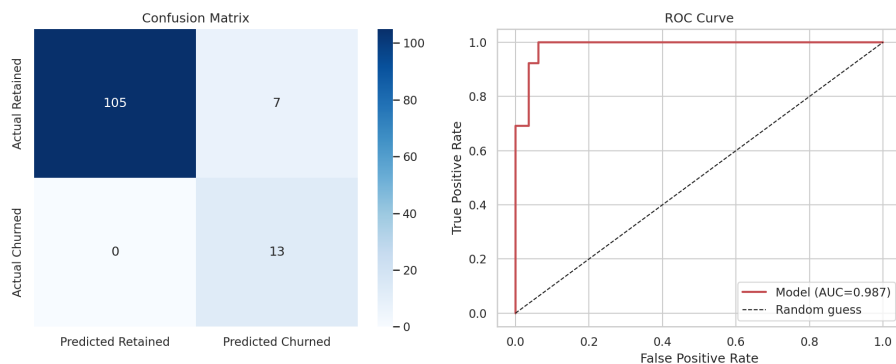


Figure 6. Confusion matrix and ROC curve (test set).

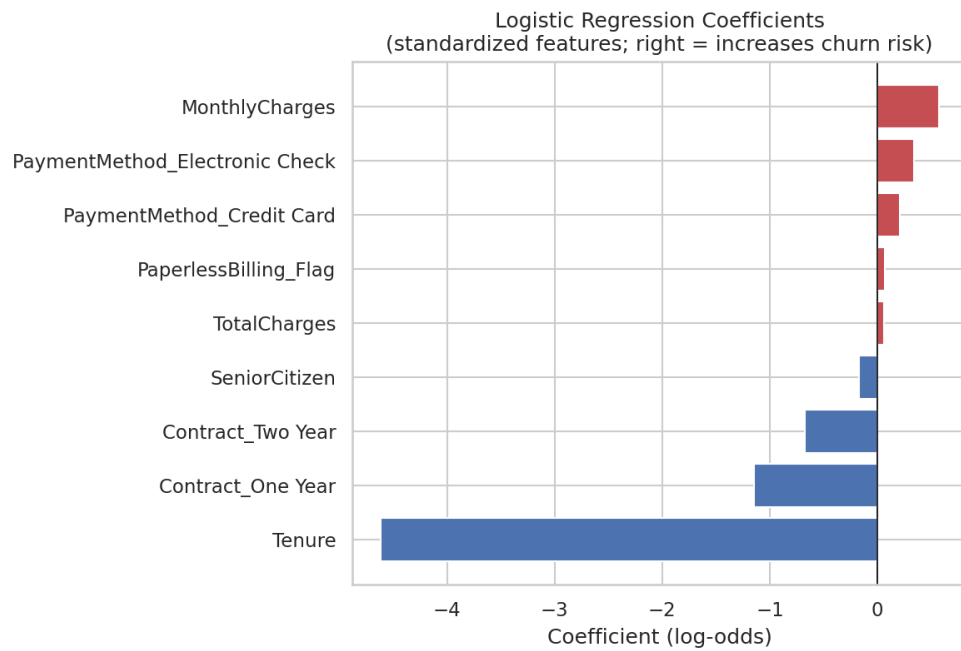


Figure 7. Standardized logistic regression coefficients.

A critical check: is a complex model necessary?

Since 100% of churners have tenure ≤ 11 months, a one-line rule was tested against the full model: flag every customer with Tenure < 12 months. Evaluated across the full dataset, this rule achieves 94.0% accuracy, 63.9% precision, and 100% recall — matching the logistic regression's recall almost exactly (the comparison is directional, not strictly apples-to-apples, since the model's figures are from a held-out test set and the rule's are from the full dataset, but the conclusion holds either way). **The honest recommendation is to ship the simple rule immediately** while the fuller model is validated on more data over time — the 9-feature model adds only modest additional ranking value within the already-identified at-risk group.

6. Customer Segmentation

K-Means clustering (k=3, chosen for interpretability; silhouette scores were modest across k=2-6, ranging 0.25-0.31, indicating this data doesn't have sharply separated natural clusters) grouped customers by Tenure, MonthlyCharges, and TotalCharges.

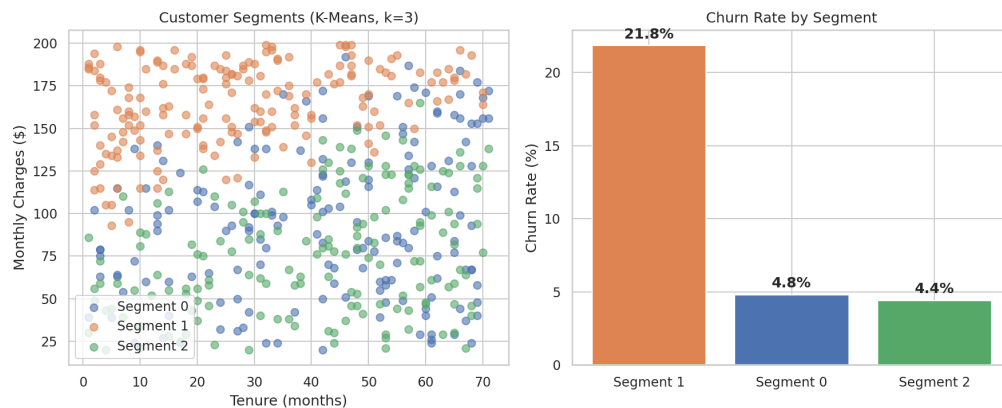


Figure 8. Customer segments and churn rate by segment.

One segment — higher monthly charges (~\$165), lower average tenure (~27 months) — carries a 21.8% churn rate, roughly 5x the other two segments (~4.5% each). This segment is the clearest target for a proactive, segment-level retention campaign, complementing the model's reactive, per-customer flagging.

7. Business Implementation Plan

Phase	Action	Owner	Success Metric
1 (Wks 1-2)	Deploy "Tenure<12mo" flag in CRM	Retention Ops	100% of eligible customers flagged in week 1
2 (Wks 3-6)	Incentivize at-risk customers onto annual contracts	Marketing/Lifecycle	Conversion rate of flagged customers
3 (Wks 5-10)	Dedicated onboarding track for high-risk segment	Customer Success	New cohort churn vs. 22% baseline
4 (Ongoing)	Retrain model quarterly	Data/Analytics	Recall stays ≥ 0.90 each refresh

Overall program success metric: reduce the 0-12-month cohort's churn rate from 63% toward parity with the rest of the base (approaching 0%) within two quarters of full rollout.

8. Testing Evidence & Limitations

Testing Evidence

- All 39 notebook cells execute top-to-bottom with zero errors (verified by parsing the saved notebook JSON for error outputs).
- Data quality was checked programmatically: missing values, duplicate rows, duplicate IDs, implausible ranges, and categorical label consistency all passed before any modeling began.
- Levene's test validated the equal-variance assumption before each independent t-test.
- The model was evaluated on a held-out test set (stratified split) never seen during training, and cross-checked against a simple, independent baseline rule to sanity-check the result rather than trusting the model's metrics in isolation.
- K-Means cluster count was chosen using silhouette scores across $k=2-6$, reported honestly as modest rather than overclaiming sharp separation.

Limitations

- Sample size (500 customers, 53 churned) limits statistical power for subgroup analyses (e.g., payment method within a single contract type).
- The extremely clean tenure/churn split (100% of churn under 12 months) may reflect how this dataset was constructed; real-world churn timing patterns should be re-validated as new data arrives rather than assumed to hold.
- The model was not validated against a true future time period (no out-of-time test set was available) — recommend monitoring performance closely after deployment.
- Clustering silhouette scores (0.25-0.31) indicate soft, not sharp, cluster boundaries; segment assignments should inform prioritization, not rigid customer categorization.